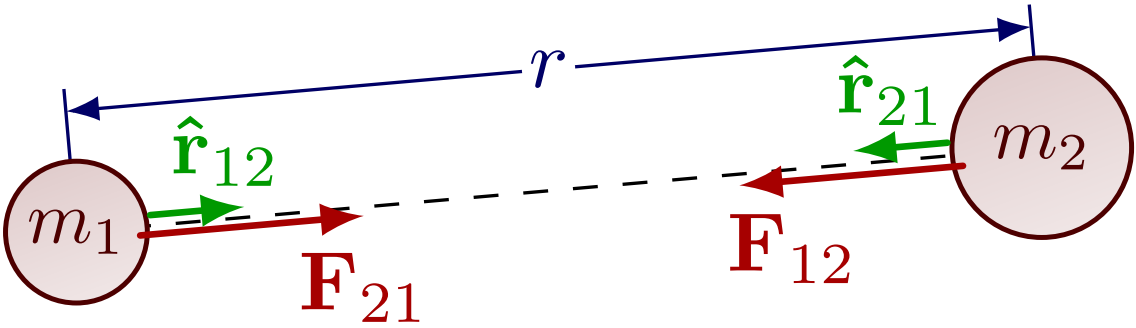
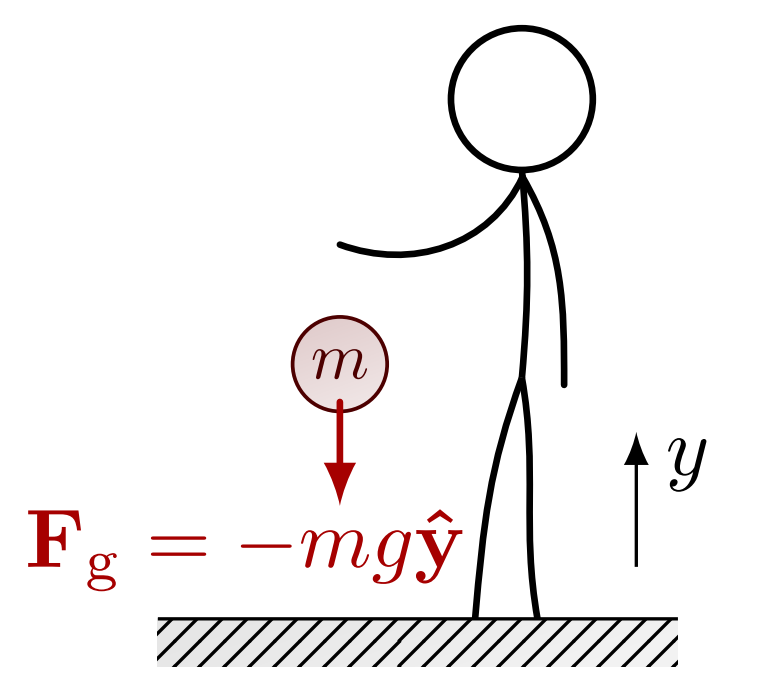
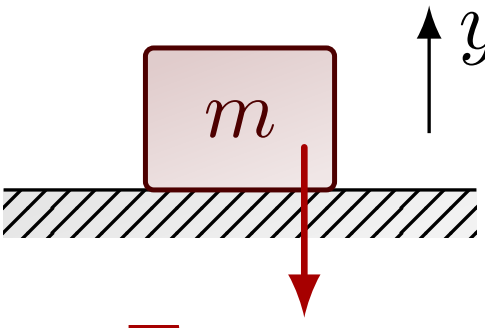
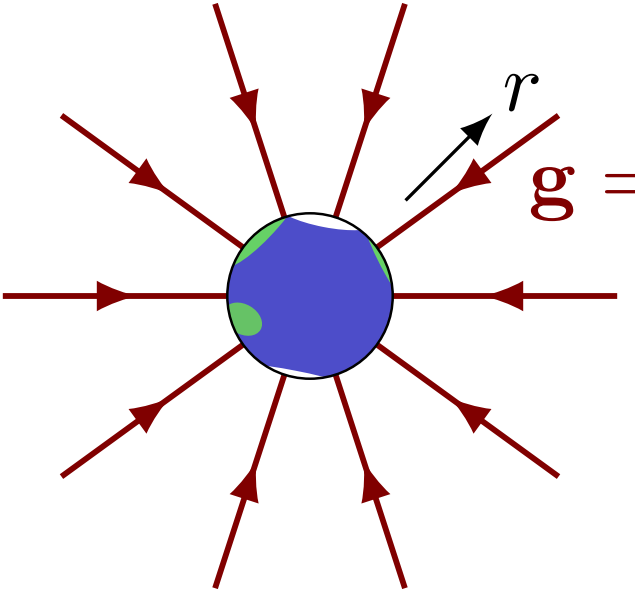

r
$\hat{\mathbf{r}}_{12}$
$\hat{\mathbf{r}}_{21}$
m_1
m_2
$\mathbf{F}_{21}$
$\mathbf{F}_{12}$

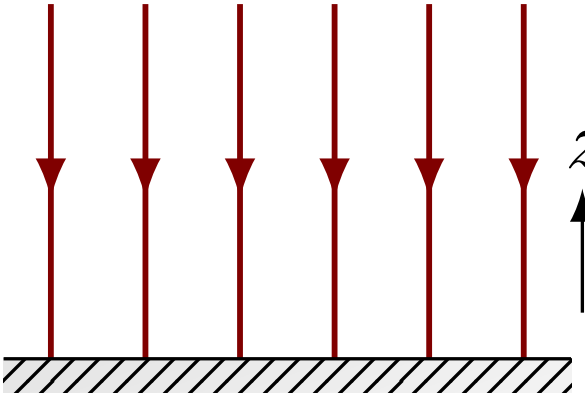

m
y
$\mathbf{F}_{\mathrm{g}} = -mg\hat{\mathbf{y}}$



$$\mathbf{F}_{\mathrm{g}} = -mg\hat{\mathbf{y}}$$



$$\mathbf{g} = -\frac{gm}{r^2}\hat{\mathbf{r}}$$


$\mathbf{g} = -g\hat{\mathbf{z}}$
z